



Molecularly imprinted electrospun nanofibre membrane assisted stir bar sorptive extraction for trace analysis of sulfonamides from animal feeds



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ABSTRACT

The widespread addition of sulfa antibiotics in feeds has long been an environment safety issue. Therefore, it is quite meaningful to constantly propose better solutions for determining the sulfa antibiotics in the environment. Herein, molecularly imprinted electrospun nanofibre membrane (MIM) assisted stir bar sorptive extraction method for the determination sulfonamides (SAs) in feed was developed. In this study, molecularly imprinted polymers (MIPs) were firstly prepared via emulsion polymerization using sulfamonomethoxine as template. Then, the morphology-controllable MIM was obtained with MIPs-doping electro-spinning solution using electro-spinning technique. Next, MIM are coated on the magnetic stir bar by heat-sealing. Finally, MIM coated stir bar was used for selective enrichment of SAs. The obtained MIM exhibited excellent class selectivity towards SAs (the selectivity factor β is 2.3–2.7). The MIM coated stir bar can be directly applied to extract SAs from feed samples. Under the optimum extraction conditions, the detection limit of the method could reach 1.5–3.4 ng/g. And satisfactory recoveries ranging from $80 \pm 6\%$ to $89 \pm 7\%$ were achieved under all three spiked levels for determination of four SAs in five different feed samples. The present work not only offers insights to develop magnetic stir bar using imprinted membrane, but also provides a new method for SAs extraction.

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Introduction

Sulfonamides (SAs) are a class of antibacterial drugs that have been extensively added in animal feeds to prevent the animal diseases and improve the feed conversion rate [1,2]. However, indiscriminate use of SAs would result in excessive accumulation in edible animal products including tissues, organs and eggs, which could indirectly affect human health since long-term use of animal products containing excessive amounts of SAs might cause drug resistance and even agranulocytosis, etc. [3–6]. Therefore, the European Union required a maximum limit for SAs residue at the total level of 100 ng/g in animal feeds. Consequently, it is of great importance to continuously develop a faster and more accurate method for the monitoring and determination of trace levels of SAs in animal feeds. So far, a variety of analytical methods have been published for the determination of SAs using high performance liquid chromatography (HPLC) coupled with various detectors,

such as ultraviolet [7,8], diode array detector [9], mass spectrometry (MS) [10,11] and Q-TOF-MS [12]. Among these approaches mentioned above, high performance liquid chromatography-tandem mass spectrometry (LC-MS/MS) is the most selective and sensitive technique for the identification and quantification of SAs [13–16].

Owing to the relatively low concentrations of SAs and the inherent complexity of the feed samples, sample pretreatment prior to the analysis is significantly necessary for the reliable determination of these compounds. Various extraction methods including ultrasonic assisted extraction [14], pressurized solvent extraction [17] and solid phase extraction (SPE) [9,18] have been widely exploited for the extraction of SAs from animal feeds. However, certain deficiencies of these methods have been revealed, such as requirement of large amount of organic solvents and lack of selectivity. As one of the mostly used solid phase micro extraction techniques, stir bar sorption extraction (SBSE) has been extensively applied in the pretreatment of complex samples. In the process of SBSE, the extraction, clean-up and enrichment procedures could be carried out in a single step, which greatly simplified the experimental procedures [19,20]. Meanwhile,

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compared with the conventional extraction methods, SBSE could achieve the increased adsorption capacity and extraction efficiency under the condition of using less organic solvents and adsorbents, indicating environmental-friendly property.

Traditional adsorbents such as polydimethyl siloxane [21–23], activated carbons [24,25], styrene-divinylbenzene [26], ethyl propiolate [27], polyacrylate, and polyethylene glycol-modified silicone [23] were commonly used in SBSE procedure. Although these adsorbents can solve the deficiency of traditional SPE and improve the analysis efficiency to some extent, the problem of selectivity in SAs detection in complex matrices still needs to be solved. In recent years, as a new type of adsorbent, molecularly imprinted polymers [28–32] have been widely used in the adsorption of pollutants with high adsorption selectivity thanks to their structure predictability, recognition specificity and application universality. We envisioned that the combination of molecular imprinting technology and modern extraction technology will inevitably improve the separation selectivity. A novel pretreatment method based on mixed matrix membranes combining molecularly imprinted technology, that is, molecularly imprinted membrane (MIM) [33–38], has been developed. In general, homogenous imprinted particles were usually treated as fillers which are well-dispersed in a polymeric matrix material. And MIM could be prepared by different methods including electrochemical method [39], photo-initiated polymerization, filling method [40] and electro-spinning [41–44]. Particularly, electro-spinning technique, which combined of various virtues, such as simple, low-cost, and controllable morphology of the MIM, has become the most promising candidate for MIM preparation.

In this study, a SBSE method using MIM as extraction material that prepared by electro-spinning technique has been developed for the fast determination of SAs in animal feed samples. The molecularly imprinted polymers (MIPs) was firstly fabricated by emulsion polymerization using sulfamonomethoxine, methacrylic acid and ethylene glycol dimethacrylate as template, functional monomers and cross-linker, respectively. The MIM was then prepared by doping certain amount of MIPs solution in the electro-spinning solvent, thus the MIP-uniformly distributed nanofiber membrane could be obtained after electro-spinning procedure. The prepared MIM was coated on the surface of the magnetic stir bar, which can be directly used to extract SAs from feed samples. And the obtained MIM coated stir bar showed satisfactory regeneration capability and excellent reusability. Combinatorial SBSE method based on the MIM possesses supplementary merits

since the proposed method not only exhibited both super high extraction efficiency and excellent adsorbing selectivity toward the analytes, but also greatly simplified the sample pretreatment procedures and decreased the matrix effect in HPLC-MS/MS analysis.

Experimental section

Chemicals and instruments

Chemicals and instruments information can be found in Supporting Information.

Preparation of the MIM

The MIM was prepared by a three-step method, and the morphology of the MIM was controllable by adjusting the preparation method. Therefore, in order to achieve the best results, polymerization conditions including amounts of the template, monomer and cross-linking agent, polymerization temperature, rotate speed, polymerization time, and electro-spinning conditions such as composition of the electro-spinning solution, DC voltage, feed rate and distance between the needle tip of the syringe and the collector were evaluated. The subsequent experiments were carried out under the optimized conditions.

Synthesis of the molecularly imprinted polymers (MIPs): The MIPs were prepared as follows: SMM (0.5 mmol) and MAA (2.0 mmol) were dissolved in 10.0 mL of DMSO. The solution was treated by ultrasound for 10 min for self-assembling. After that, EGDMA (8.0 mmol), styrene (1.0 mL) and AIBN (0.05 g) were added. The mixture was dispersed into 100 mL of 0.25% SDS aqueous solution, followed by adding 0.8 g of octadecanol as stabilizer. The mixture was then intensively sonicated for 20 min in ice bath and the obtained emulsion was polymerized at 60 °C for 24 h under nitrogen atmosphere at 400 rpm/min. The produced polymers were centrifuged and washed with ethanol. After removing SMM templates with methanol/acetic acid (8:2, v/v) by soxhlet extraction with 24 h, MIPs were obtained. Non-imprinted polymers (NIPs) were prepared and processed in the same manner as MIPs, except that no template (SMM) was added during polymerization.

Preparation of the electro-spinning solution: 3 g of PAN was completely dissolved into 40.0 mL of DMF by stirring at 40 °C for 4 h, then 0.4 g of the polymers (MIPs or NIPs) was added and the

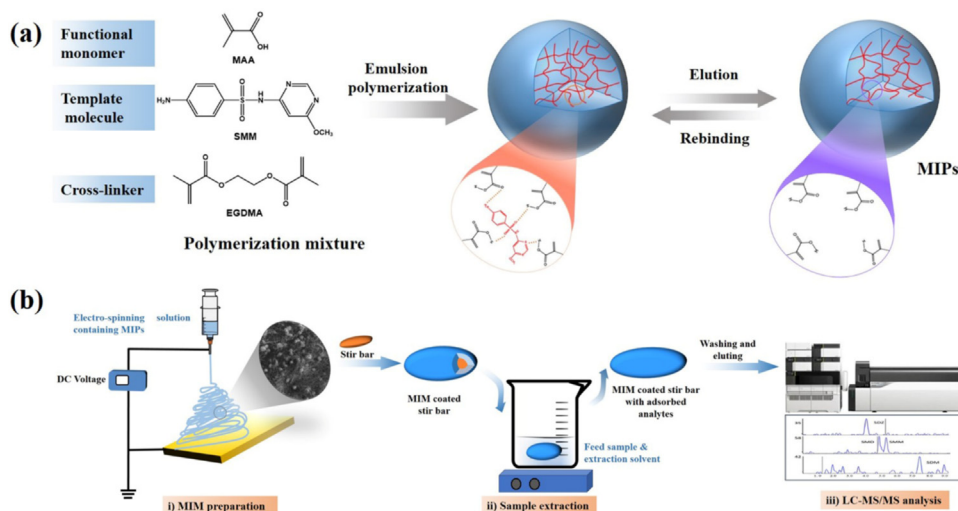


Fig. 1. (a) Schematic illustration of the synthetic procedure for MIPs. (b) The procedure for MIM preparation and the process for the detection of trace SAs in feed samples.

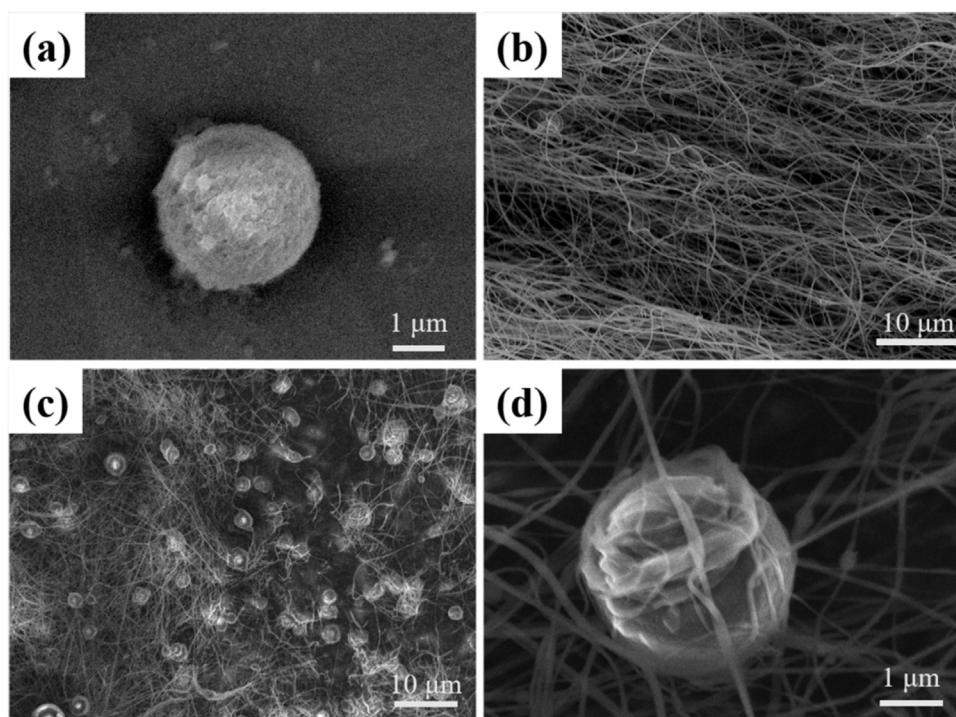


Fig. 2. SEM images of (a) MIPs, (b) PAN fibers and (c, d) MIM.

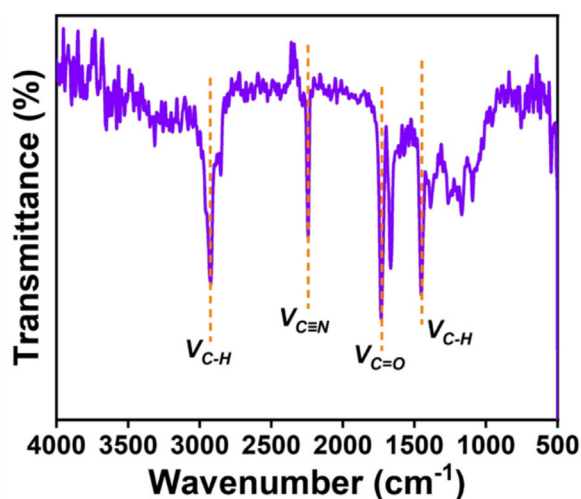


Fig. 3. FT-IR spectrum of MIM.

mixture was continuously stirred until the system became uniform.

Electro-spinning: Preparation of the MIM was accomplished by electro-spinning of the MIPs. The freshly prepared electro-spinning solution was loaded into the syringe, 18 kV of DC voltage was applied, the feed rate of the solution was 0.3 mL/h, distance between the needle tip of the syringe and the collector was 15 cm, the obtained MIM nanofibers were finally peeled off from the collector and dried at 80 °C for 6 h and stored in hermetic bag for further use.

The non-molecularly imprinted electrospun nanofibre membrane (NIM) was prepared the same way except for NIPs were doped into the electro-spinning solution instead of MIPs.

Binding experiment

To evaluate the adsorption performance of the MIM, static adsorption experiment was carried out in methanol aqueous solution (v/v, 50/50). MIM or NIM (10.0 mg) was dispersed into SMM solutions (1.0 mL) with various concentrations (0.25–2.0 mmol/L), respectively. The mixed solutions were shaken vigorously for 24 h at 25 °C, and the suspensions were then filtered and analyzed by HPLC–MS/MS. The amount of SMM bounded to the polymers was obtained by subtracting the free concentration from initial concentration of SMM.

The selectivity of the prepared MIM and NIM were assessed with template molecule (SMD), structural analogues (SDZ, SMD, SDM), and reference compounds (aniline). The experiment was carried out by adding 10.0 mg MIM or NIM into 1.0 mL of each standard solution at the concentration of 0.1 mmol/L, respectively. After shaking the mixtures for 24 h at 25 °C, the supernatants were separated and analyzed by HPLC–MS/MS.

Sample preparation

Firstly, the MIM-stir bar was made by packing 150 mg of the MIM outside a stir bar, after sealing both sides by heating, the MIM-stir bar was soaked in ethyl acetate. Then, 0.5 g of feed sample was mixed with 10.0 mL of ethyl acetate and sonicated for 10 min before the MIM-stir bar was added. The extraction was performed by stirring the mixture for 25 min, afterwards, the MIM-stir bar with absorbed analytes was washed with 4.0 mL of 50% methanol aqueous solution, and subsequently eluted by 2.0 mL of methanol solution containing 5% acetic acid under ultrasound (5 min). The collected eluate was dried by nitrogen and re-dissolved with 1.0 mL of 30% methanol aqueous solution and filtered through a 0.45 μm membrane prior to LC–MS/MS analysis.

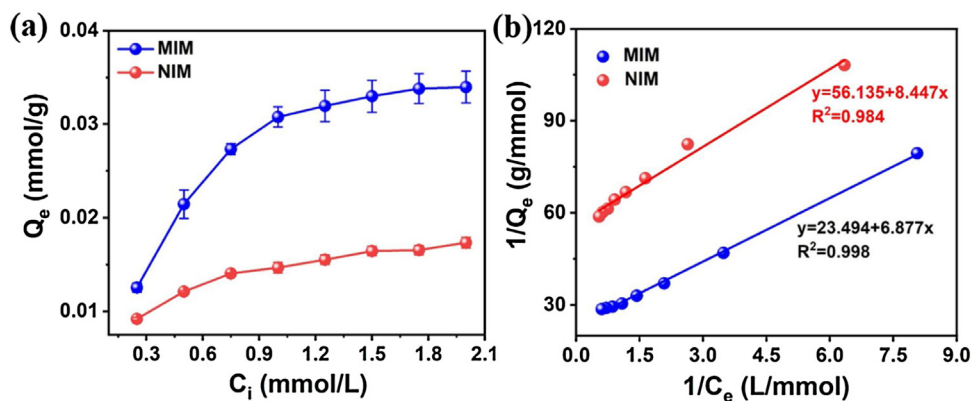


Fig. 4. (a) Binding isotherms of MIM and NIM toward SMM. (b) The Langmuir adsorption isotherm fit of SMM.

Table 1

The equations and parameters of Langmuir and Freundlich isotherm model.

Adsorbents	Langmuir			Freundlich		
	$Q_e = Q_m K_L C_e / (1 + K_L C_e)$			$\ln Q_e = \ln K_F + \frac{1}{n_F} \ln C_e$		
	Q_m (mmol/g)	K_L (L/mmol)	R^2	$\ln K_F$ (mmol/g)	$1/n_F$	R^2
MMIPs	0.0426	3.416	0.998	−3.438	0.385	0.920
MNIPs	0.0178	6.646	0.984	−4.189	0.246	0.975

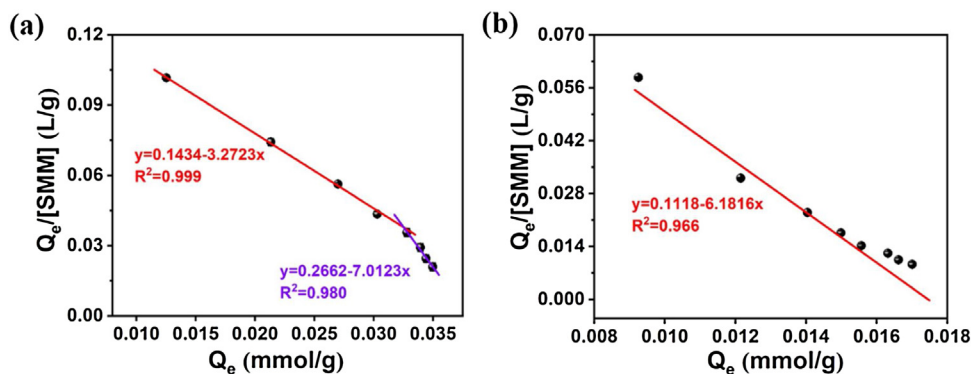


Fig. 5. Scatchard plots of (a) MIM and (b) NIM.

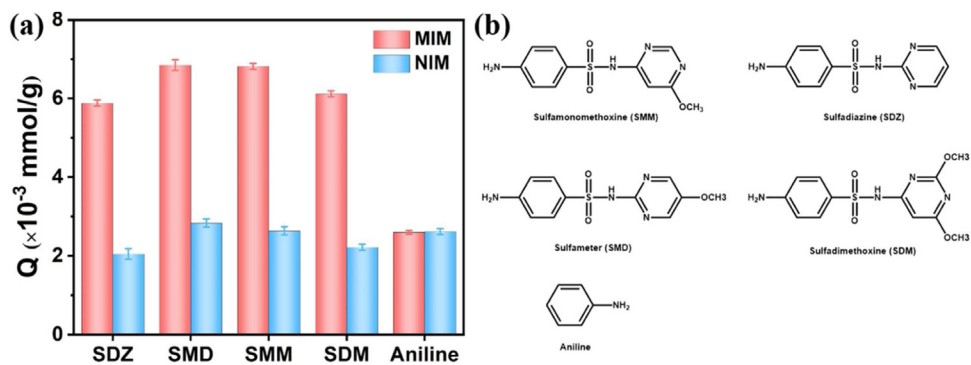


Fig. 6. (a) Binding capacities of sulfadiazine (SDZ), sulfameter (SMD), sulfamonomethoxine (SMM), sulfadimethoxine (SDM) and aniline on MIM and NIM. (b) Chemical structure of SDZ, SMD, SMM, SDM and aniline.

Sample preparation for traditional SPE method

Determination method for SAs in feed samples by current Chinese National Standard (GB/T 19542–2007) was used for comparison. The sample preparation procedures were as follows: 2.0 g feed samples were mixed with 50.0 mL ACN and the mixture were vibrated for 30 min, when the extraction completed, the solvent were separated and slowly loaded onto a cartridge packed with Alumina-B (previously conditioned with ACN) before the cartridge was dried. Afterwards, 2×2.0 mL ACN-water (25:75, v/v) containing 0.3% acetic acid was used to elute the target SAs. Finally, the eluate was collected and filtered through a 0.45 μm membrane prior to analysis.

Results and discussion

Preparation and characterization of MIM

The process of MIPs and MIM preparation and our strategy for the determination of SAs was depicted in Fig. 1. The procedure of MIM fabrication can be summarized by the following three steps. Firstly, the MIPs was synthesized via a one-pot emulsion polymerization (Fig. 1a). MAA was used as functional monomer, which could interact with template SMM by hydrogen bonding. The emulsion polymerization method used not only endowed the MIPs with uniform structure, but also was conducive to embedding the MIPs into the subsequent polymer spinning materials. After the removal of the templates, the recognition sites in MIPs complementary to SAs in shape, size, and functional group were formed, leading to high rebinding capacity of MIPs to SAs. Secondly, electro-spinning solution was obtained by doping certain amount of MIPs into PAN solution. Compared with polystyrene, PAN as matrix material endowed imprinted electro-spun nanofiber with higher strength. Thirdly, preparation of the MIM was accomplished by electro-spinning of the MIPs-doping PAN solution. The imprinted cavities in MIPs for SAs enabled the MIM with the selective adsorption for SAs.

The morphologies of the prepared MIPs, PAN nanofibers and MIM were observed by FE-SEM. As shown in Fig. 2a, the as-synthesized MIPs presented a regular shape and exhibited spherical structure with a rough surface. For PAN nanofibers (Fig. 2b), it can be clearly seen that a large amount of uniform nanofibers. Fig. 2c and 2d showed that the spherical MIPs were well dispersed in the net structures of the PAN nanofibers.

Then the functional groups of the MIM were characterized by FT-IR. As illustrated in Fig. 3, two peaks at 2241 cm^{-1} (C–N stretching), 2920 cm^{-1} (C–H bending vibration) and 1453 cm^{-1} (C–H stretching) indicated the existence of PAN. And peaks near 1731 and 1668 cm^{-1} represented the stretch of C=O, which verified the MIPs were successfully doped into the nanofibers.

Binding characteristics of MIM

The recognition ability of the MIM toward the template SMM was investigated. The adsorption isotherm was performed at 25°C ,

Table 2
Imprinting and selectivity factors of MIM.

Analytes	$Q (\times 10^{-3} \text{ mmol/g})$		α	β
	MIM	NIM		
SDZ	5.9	2.0	3.0	2.3
SMD	6.9	2.8	2.5	2.7
SMM	6.8	2.6	2.6	2.6
SDM	6.1	2.2	2.8	2.3
Aniline	2.6	2.6	1.0	–

and the adsorption amount of SMM was calculated using the following equation (Eq. 1):

$$Q_e = (C_i - C_e) V / m \quad (1)$$

Where Q_e (mmol/g) is the amount of SMM adsorbed on the MIM at equilibrium; C_i and C_e (mmol/mL) are the initial and equilibrium concentrations of SMM, respectively; V (mL) is the solution volume; m (g) is the weight of the MIM.

As shown in the binding isotherms (Fig. 4a), with the increasing of initial concentration of SMM, the amount of SMM bound to the MIM at equilibrium increased. For comparison, the adsorption experiments of NIM were also performed. The results demonstrated that only a small amount of SMM was bound to NIM, which attributed to the surface nonspecific adsorption. And obviously, MIM exhibited much higher recognition ability than NIM among all concentration ranges. The above data suggested the successful construction of template recognition sites in MIM, which could rebind the templates with high adsorption capacity. By contrast, there are no imprinted cavities in NIM, thus resulting in its low adsorption capacity for SMM. In addition, Langmuir and Freundlich isotherm adsorption model were used for fitting the adsorption data for further investigation on the adsorption mechanism of MIM and NIM. The optimum fitting plots with the Langmuir isotherm models were presented in Fig. 4b. And the equations and parameters of Langmuir and Freundlich isotherm model are shown in Table 1. Through comparison of correlation coefficient and linearity data, the Langmuir isotherm model was found to better fit the adsorption experiment data of MIM, which indicated that the adsorption process of SMM by MIM can be regarded as a monolayer adsorption process.

The binding data were further processed by Scatchard analysis to evaluate the binding properties of MIM and NIM, and the Scatchard plots were constructed according to the following equation (Eq. 2):

$$Q_e / [SMM] = (Q_{\max} - Q_e) / K_d \quad (2)$$

Where $[SMM]$ (mmol/L) is the free SMM concentration at adsorption equilibrium; K_d is the dissociation constant and the $Q_{\max}$ (mmol/g) is the apparent saturated adsorption capacity. The values of K_d and $Q_{\max}$ can be calculated from the slope and intercept of the linear line plotted in $Q_e/[SMM]$ versus Q_e .

As shown in Fig. 5a, the Scatchard plot for MIM consisted of two linear curves with different slopes, which suggested two different types of binding sites on the MIM, including high-affinity sites for specific binding toward SAs and low-affinity sites with no selectivity. K_d and $Q_{\max}$ of the high-affinity sites were obtained by the left part of the Scatchard plot ($Q_e/[SMM] = -3.27Q_e + 0.14$) and were calculated to be $305.8\text{ }\mu\text{mol/L}$ and $42.8\text{ }\mu\text{mol/g}$, respectively. While the K_d and $Q_{\max}$ of the low-affinity sites were calculated to be $142.7\text{ }\mu\text{mol/L}$ and $37.9\text{ }\mu\text{mol/g}$, respectively, based on the right part of the Scatchard plot ($Q_e/[SMM] = -7.0Q_e + 0.270$).

On the other hand, the Scatchard plot for NIM revealed simple linearity, which indicated homogeneous binding sites on the NIM, and the values of K_d and $Q_{\max}$ were $161.8\text{ }\mu\text{mol/L}$ and $18.1\text{ }\mu\text{mol/g}$, respectively (Fig. 5b).

Selectivity performance of MIM

Selectivity of the MIM was evaluated by the comparison of binding capacities between MIM and NIM toward SAs and reference compound (Aniline). As can be seen from Fig. 6, the amounts of SMM together with its structural analogues bound to the MIM were much higher than that of aniline, which verified the high selectivity of MIM towards SAs. On the other hand, no obvious difference was shown between the binding amounts of SAs and

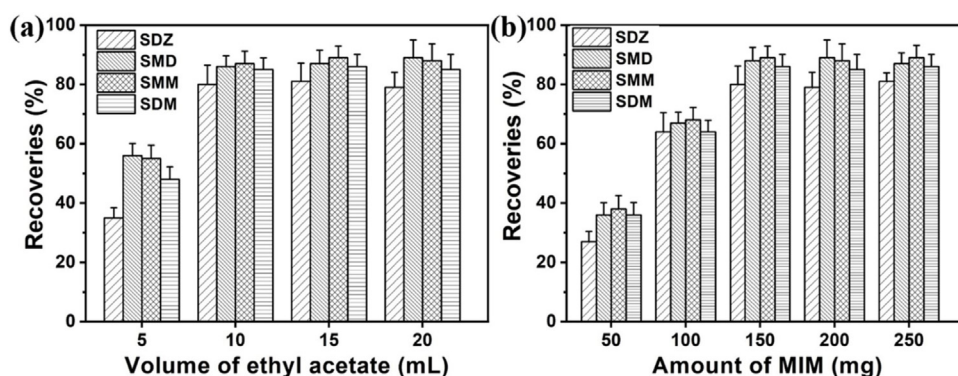


Fig. 7. Effect of ethyl acetate volume and MIM amount on the recoveries of SAs.

Table 3

Validation of the method.

Analytes	Linearity range (ng/g)	Calibration curve	R ²	LOD (ng/g)	LOQ (ng/g)
SDZ	10–1000	Y = 132 + 31X	0.998	3.4	9.6
SMD	10–1000	Y = -14 + 54X	0.997	1.5	4.6
SMM	10–1000	Y = 35 + 56X	0.998	2.6	7.5
SDM	10–1000	Y = 41 + 50X	0.999	3.1	8.5

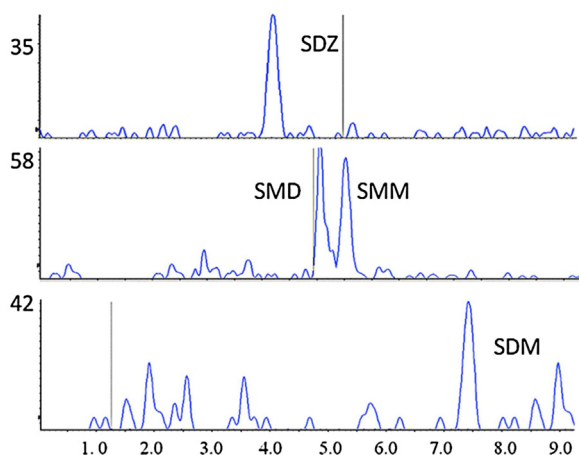


Fig. 8. LC-MS/MS chromatograms of spiked feed sample (10 ng/g).

aniline for NIM, and the adsorption amounts of the SAs were much lower than those bound to the MIM. The weak adsorption of SAs and the reference compound should be attributed to the nonspecific adsorption. The above results implied that MIM possessed both high adsorption capacity and excellent selectivity toward SAs than NIM. MIM containing plenty of MIPs has a large amount of imprinted sites, which could match the size, shape and functional groups of SAs, resulting in excellent rebinding behavior

toward the SAs. As for the reference compound, the recognition sites cannot bind them tightly due to the mismatched structure.

In addition, the imprinting factor (α) and selectivity factor (β) of MIM were studied based on the obtained data, according to the following equations (Eq. 3 and Eq. 4):

$$\alpha = Q_{MIM} / Q_{NIM} \quad (3)$$

$$\beta = \alpha_{SAS} / \alpha_{Aniline} \quad (4)$$

Where Q (mmol/g) is the adsorption amount of the SAs, α is the imprinting factor, which represents the selectivity of MIM towards SAs compared to that of NIM; while β is the selectivity factor that corresponds to the selectivity of MIM towards SAs and the reference compound.

The resulting data were showed in Table 2. The selectivity of MIM towards SAs was very satisfactory with high imprinting factor and selectivity factor of 2.5–3.0 and 2.3–2.7, respectively.

Optimization of the extraction conditions

Extraction parameters that could affect the performance of the extraction, such as extraction solvent, amount of MIM, extraction time, washing and desorption conditions were optimized. When one parameter was under investigated, the other parameters were fixed at their optimal values. All the following optimization experiments were performed in triplicate by analyzing spiked feed samples (50 ng/g). The optimum extraction conditions were as

Table 4

The intra-day precisions and recoveries of the assay (n = 6).

Analytes	10 ng/g		25 ng/g		100 ng/g	
	Recovery (%)	RSD (%)	Recovery (%)	RSD (%)	Recovery (%)	RSD (%)
SDZ	75	4.5	81	6.5	81	3.8
SMD	88	3.4	89	3.7	90	5.4
SMM	87	5.2	89	4.5	89	3.7
SDM	86	3.7	85	5.4	86	4.1

Table 5

The inter-day precisions and recoveries of the assay (n = 6).

Analytes	10 ng g ⁻¹		25 ng g ⁻¹		100 ng g ⁻¹	
	Recovery (%)	RSD (%)	Recovery (%)	RSD (%)	Recovery (%)	RSD (%)
SDZ	77	3.6	81	3.9	80	7.9
SMD	86	5.1	89	4.6	88	6.4
SMM	87	2.9	89	5.7	87	2.8
SDM	82	3.6	82	3.7	82	3.7

Table 6

Recoveries of SAs in different feed samples (mean ± S.D., n = 3).

Analytes	Suckling pig feed		Chicken feed		Laying-hen feed 1		Laying-hen feed 2		Pig feed	
	Founded (ng g ⁻¹)	Recovery (%)	Founded (ng g ⁻¹)	Recovery (%)	Founded (ng g ⁻¹)	Recovery (%)	Founded (ng g ⁻¹)	Recovery (%)	Founded (ng g ⁻¹)	Recovery (%)
SDZ	8.1 ± 0.3	81 ± 3	8.0 ± 0.6	80 ± 6	8.2 ± 0.4	82 ± 4	8.0 ± 0.2	80 ± 2	8.0 ± 0.7	80 ± 7
SMD	8.7 ± 0.4	87 ± 4	8.9 ± 0.7	89 ± 7	8.8 ± 0.4	88 ± 4	8.5 ± 0.3	85 ± 3	8.7 ± 0.5	87 ± 5
SMM	8.8 ± 0.5	88 ± 5	8.7 ± 0.4	87 ± 4	8.7 ± 0.5	87 ± 5	8.4 ± 0.6	84 ± 6	8.7 ± 0.4	87 ± 4
SDM	8.5 ± 0.3	85 ± 3	8.4 ± 0.6	84 ± 6	8.3 ± 0.7	83 ± 7	8.2 ± 0.3	82 ± 3	8.5 ± 0.4	85 ± 4

Table 7

Comparison of the proposed method with other reported methods for determination of SAs in recent years.

Method	Real sample	Linearity (μg/L)	R ²	LOD (μg/L)	Recovery (%)	RSD (%)	Ref.
MIPs-SPE-HPLC-DAD	Milk	50–500	>0.9325	1.9–13.3	85.8–115.7	0.5–13.4	[46]
MIPs-SPE-HPLC-MS/MS	Animal-derived foods and water	1–2500	0.996	0.02–0.1	63.49–115.72	1.64–4.68	[47]
MISG-SEME-HPLC	Water	1–500	0.9959–0.9999	0.06–0.17	80–96	<10	[48]
MIP-SPE-HPLC	Soil and sediment	3.0–1.0 × 10 ⁴	0.9994	1.0	95.0–105.0	<8	[49]
MIP-SPE-LC-MS/MS	Honey	10–1000	0.997–0.999	1.5–4.3	67.1–93.6	3.7–7.9	[50]
MIP-SPE-HPLC-UV	Chicken meat	0.5–150	0.992–0.995	0.1–0.5	92–106	4.3–5.4	[51]
SPE-HPLC-DAD	Milk	10–200	>0.998	1.9–13.3	85.8–115.7	2–15	[52]
MSPE-HPLC-DAD	Animal foodstuffs	1–200	0.8535	2–10	70–92	5.6–13	[53]
MIM-SBSE-HPLC-MS/MS	Animal feeds	10–1000	0.997–0.999	1.5–3.4	80–89	2.8–7.9	Present work

follows (Fig. 7): 10.0 mL of ethyl acetate was chosen as extraction solvent. 150 mg of MIM were used as the optimal adsorbents amount. 25 min of stirring was sufficient. 4.0 mL methanol-water (50:50, v/v) was used as the washing solution. The MIM with captured SAs were subjected to ultrasound for 2 min during the washing process. 2.0 mL of methanol-acetic acid (95:5, v/v) was regarded as the optimal elution conditions.

Evaluation of the method

Matrix effect (signal suppression or enhancement) is a well-known problem that could greatly influence the accuracy of LC-MS/MS [45]. Because the co-extractives of complex samples can greatly affect the signals of analytes or enhance the background noise, that may compromise the quantitative of the analytes at trace levels.

The matrix effect of the proposed method was calculated according to Equation 5 (Eq. 5), through comparing the calibration curves obtained by high-purity water (peak area A) and by blank feed sample extract (peak area B) in the concentration range of 10–1000 ng/g:

$$\text{Matrix effect (\%)} = [B/A - 1] \times 100 \% \quad (5)$$

The results showed slight signal suppressions (8.7–12.8%) for the four SAs, which were much lower than the matrix effect (17.9–38.1%) obtained by the traditional SPE method, and this result further corroborated the higher selectivity of the MIM than traditional SPE adsorbents. Moreover, no significant difference was observed between different feed samples, which meant the matrix effect of the proposed method between different samples was

negligible. In order to achieve more reliable results, matrix-matched calibration curves were used for quantification in this work.

The linear range of the calibration curves obtained by analyzing spiked feed samples was 10–1000 ng/g, limit of detection (LOD, signal/noise ratio of 3:1) and limit of quantification (LOQ, signal/noise ratio of 10:1) of the proposed method for SAs were in the range of 1.5–3.4 ng/g and 4.6–9.6 ng/g, respectively (Table 3). LC-MS/MS extracted ion chromatograms of spiked feed samples (10 ng/g) are illustrated in Fig. 8.

Precision was evaluated by measuring intra- and inter-day RSDs by analyzing spiked feed samples at concentrations of 10, 25, 100 ng/g six times in one day and in six consecutive days, respectively. The results are shown in Table 4 and Table 5, the recoveries of the SAs were in the range of 72–90% at all fortified levels.

Application of the proposed method

In order to validate the applicability of the proposed method, it was used for the determination of SAs in five different feed samples including regular chicken feed, pig feed, suckling pig feed and two types of laying-hen feeds. No SAs was detected from the original feed samples. To further evaluate the accuracy of the proposed method, the recovery test was performed by spiking the feed samples with SA standards at the concentration of 10 ng/g. Satisfactory results of the recoveries ranging from 80 ± 6%–89 ± 7% were achieved (Table 6), which were much higher than those obtained by traditional SPE method (65 ± 7%–80 ± 6%) mentioned in 'Section 2.5'. Moreover, the analytical procedure of

the proposed method was much more efficient due to employment of the SBSE and high-selectivity adsorbent. We compared the method proposed in this study with the reported methods based on molecularly imprinting technique for determination of SAs, and the results are summarized in Table 7. The data showed that the presented method possessed wide linear range, low detection limit and relative standard deviation comparable to other methods, indicating the excellent applicability of the method.

Conclusion

In this paper, MIM doped with MIPs exhibited high selectivity toward SAs was successfully prepared by electro-spinning technique and applied as SBSE adsorbent for the extraction of SAs from feed samples, followed by HPLC-MS/MS analysis. The morphology and adsorption properties of the MIM were comprehensively characterized and satisfactory analytical performances were achieved. Compared to traditional SPE methods, the combination of SBSE and molecular imprinting technique presented many advantages, including simpler sample preparation procedure, less solvent consumption and higher extraction efficiency. Moreover, the reusability and stability of MIM are also satisfactory. Therefore, this method could be considered of great potential in rapid separation and enrichment of trace pollutant from different complex samples.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jiec.2021.01.045>.

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